



UNIVERSITY OF GENEVA
Department of Informatics

CC1

Digital Forensics

14x065

Michel Jean Joseph Donnet

October 29, 2025

Table of Contents

1	Digital Forensics: Principles and Techniques	2
1.1	Principles of digital forensics	2
1.2	Digital multimedia forensics: definition, classification	2
1.2.1	Core Principle	2
1.3	Active and passive approaches for multimedia authentication	3
1.3.1	Active approach	3
1.3.2	Passive approach: multimedia forensics	3
1.4	Image generation pipeline	3
1.5	Camera identification based on photo response non-uniformity (PRNU) .	4
1.6	Media forensics based on in-camera fingerprints	4
1.6.1	Source identification	4
1.6.2	Fingerprint applications	4
1.7	Application of geometrical fingerprints in digital forensics	4
1.8	Out-camera fingerprints in digital forensics	5
1.9	Principles of image JPEG compression	5
1.10	JPEG compression: general compression scheme	5
1.11	JPEG compression: sources of redundancies	5
1.12	JPEG compression: luminance/chrominance processing	6
1.13	JPEG compression artefacts	6
1.14	Detection of double JPEG compression	6
1.15	Image editing forensics: inconsistency in light	6
1.16	Image editing forensics: local filtering traces	6
1.17	Photo manipulation with inconsistent shadows	7
1.17.1	Basic assumptions	7
1.17.2	Cast shadow analysis	7
1.17.3	Attached shadow analysis	7
1.17.4	Mathematical model	7
1.18	Traces of image editing: resampling detection	7
1.19	Traces of image editing: contrast enhancement detection	8
1.19.1	Global contrast enhancement	8
1.19.2	Local contrast enhancement	8

1 Digital Forensics: Principles and Techniques

1.1 Principles of digital forensics

Core Concept: A systematic methodology for investigating digital evidence while maintaining legal integrity and chain of custody.

Four Key Phases:

1. **Acquisition:** Create forensic copies using write-blockers to preserve original evidence integrity
2. **Identification:** Locate, recover, and categorize relevant digital evidence using specialized tools
3. **Evaluation:** Analyze evidence to reconstruct timelines, establish causality, and validate hypotheses
4. **Presentation:** Document findings in court-admissible formats understandable to legal professionals

Applications: Criminal investigations, civil litigation, corporate internal audits, incident response

1.2 Digital multimedia forensics: definition, classification

Definition: Scientific discipline verifying authenticity and origin of digital media through analysis of inherent traces.

Media Types: Images, video, audio files

Critical Problem: Digital media can no longer be trusted as objective evidence due to: - Sophisticated editing tools enabling seamless manipulation - Social impact: erosion of trust in visual information - Scientific challenge: establishing definitive authenticity protocols

1.2.1 Core Principle

Fundamental Insight: Every digital media file contains inherent traces (“digital fingerprints”) from: - Acquisition device characteristics - Processing operations applied - Physical scene properties

Digital fingerprints Three distinct fingerprint categories:

- **In-camera fingerprints:** Hardware/software signatures unique to specific devices (lens distortions, sensor patterns, processing algorithms)
- **Out-camera fingerprints:** Statistical alterations from editing operations (compression artifacts, filter applications)
- **Scene fingerprints:** Physical consistency indicators (lighting direction, shadow coherence, perspective geometry)

1.3 Active and passive approaches for multimedia authentication

1.3.1 Active approach

Proactive Protection: Embed security features during creation phase

Methods:

1. **Cryptographic Signatures:** Hash-based authentication with potential blockchain integration
2. **Digital Watermarking:**
 - **Fragile watermarks:** Detect any modification
 - **Semi-fragile watermarks:** Allow legitimate processing while detecting malicious tampering

Disadvantages Technical Limitations: - Multimedia-specific encryption challenges: loss of scalability, high computational demands - Inability to distinguish between compression (acceptable) and tampering (malicious) - Protection ceases after decryption, requiring additional rights management

1.3.2 Passive approach: multimedia forensics

Reactive Analysis: Investigate existing media without pre-embedded markers

Forensic Questions: - **Source identification:** Which device created this media?
- **Integrity verification:** Has this media been altered from its original state?

Advantages Practical Benefits: - No requirement for pre-existing security infrastructure - Applicable to any digital media regardless of origin - Works with standard consumer devices and formats

1.4 Image generation pipeline

Process Understanding: Digital cameras apply specific processing chains that leave detectable fingerprints

Eight-Stage Pipeline:

1. **Light projection:** Optical system focuses light onto sensor array
2. **Amplification/quantization:** Convert photon energy to digital values
3. **White balancing:** Adjust color gains for proper color reproduction
4. **Color filtering:** Bayer pattern CFA separates RGB components
5. **Demosaicking:** Interpolate missing color values from neighboring pixels
6. **Color correction:** Transform to standard color spaces (sRGB, Adobe RGB)

7. **Enhancement:** Apply sharpening, noise reduction, gamma correction
8. **Compression/storage:** Encode in JPEG/HEIC formats with metadata

Forensic Value: Each stage introduces device-specific patterns usable for authentication

1.5 Camera identification based on photo response non-uniformity (PRNU)

Unique Identifier: Manufacturing imperfections in image sensors create device-specific noise patterns

Noise Components: - **Fixed Pattern Noise (FPN):** Additive noise visible in uniform dark conditions - **Photo-Response Non-Uniformity (PRNU):** Multiplicative noise from pixel sensitivity variations - the primary forensic fingerprint

Mathematical Model: $I(x, y) = I_0(x, y) + I_0(x, y) \cdot K(x, y) + N(x, y)$

Where: - $I(x, y)$: Observed pixel value - $I_0(x, y)$: Ideal noise-free image - $K(x, y)$: PRNU pattern (camera fingerprint) - $N(x, y)$: Additive noise term

Implementation: Estimate $K(x, y)$ by averaging noise residuals from multiple reference images, then correlate with questioned images

1.6 Media forensics based on in-camera fingerprints

1.6.1 Source identification

Device Attribution: Determine capture device through hierarchical analysis

Identification Levels: - Device class (CCD vs CMOS sensors) - Manufacturer and model identification - Individual unit identification

1.6.2 Fingerprint applications

Dual Purpose: - **Authentication:** Match against reference databases of known devices - **Tampering detection:** Identify fingerprint inconsistencies indicating manipulation

Specific techniques **CFA Analysis:** Detect anomalies in color filter array interpolation patterns **Processing Chain Analysis:** Identify inconsistencies in white balance, sharpening, compression parameters **Device Model Profiling:** Compare against expected behaviors for specific camera models

1.7 Application of geometrical fingerprints in digital forensics

Physical Consistency Checks: Use scene geometry properties to detect forgeries

Detection Methods:

- **Copy-move detection:** Identify duplicated regions using block-matching algorithms
- **Lighting analysis:** Verify consistency of highlights/shadows across objects
- **Perspective verification:** Check geometric coherence in scene composition
- **Chromatic aberration:** Detect inconsistencies in lens distortion patterns

Practical Application: Most effective for identifying composite images from multiple sources

1.8 Out-camera fingerprints in digital forensics

Post-processing Detection: Identify editing operations through statistical analysis

Detectable Operations: - Compression history (JPEG quantization tables) - Geometric transformations (scaling, rotation parameters) - Filter applications (convolution kernel signatures) - Contrast/color adjustments (histogram modifications)

Technical Basis: Each processing operation alters statistical properties in detectable ways

1.9 Principles of image JPEG compression

Compression Rationale: Exploit redundancies and human visual limitations

Redundancy Types: - **Spatial:** Correlation between neighboring pixels - **Spectral:** Relationships between color channels - **Psychovisual:** Information poorly perceived by humans

Key Insight: Human vision is more sensitive to luminance (brightness) than chrominance (color), enabling chroma subsampling

1.10 JPEG compression: general compression scheme

Three-Stage Process:

1. **Transform (T):** Convert spatial data to frequency domain using Discrete Cosine Transform (DCT)
2. **Quantize (Q):** Reduce precision using quality-dependent quantization matrices
3. **Encode (C):** Apply entropy coding (Huffman/RLE) for final compression

Quantization Formula: $F_{quantized}[u, v] = \text{round} \left(\frac{F[u, v]}{Q(u, v)} \right)$

Where $Q(u, v)$ represents frequency-dependent quantization steps

1.11 JPEG compression: sources of redundancies

Compression Efficiency: Different redundancy types enable various compression strategies

Redundancy Exploitation: - **Spatial:** DCT concentrates energy into few coefficients - **Spectral:** Chroma subsampling (4:2:0, 4:2:2) - **Psychovisual:** Quantize high frequencies more aggressively

1.12 JPEG compression: luminance/chrominance processing

Color Space Strategy: Convert RGB to YCbCr for separate processing

YCbCr Benefits: - Y (luminance): Preserved at full resolution for detail retention
- Cb/Cr (chrominance): Subsampled with minimal perceptual impact

Data Reduction Example: 64×64 block: $\text{RGB} = 3 \times 4096 = 12,288$ values $\rightarrow$ $\text{YCbCr } 4:2:0 = 4096 + 2 \times 1024 = 6,144$ values (50% reduction)

1.13 JPEG compression artefacts

Forensic Indicators: Compression imperfections provide authenticity clues

Visible Artifacts: - **Blocking:** 8×8 block boundary discontinuities at low quality factors - **Ringings:** Ghosting effects near strong edges from frequency truncation - **Graininess:** Quantization noise appearing as random speckling - **Blurring:** High-frequency detail loss from aggressive quantization

1.14 Detection of double JPEG compression

Manipulation Indicator: Multiple compression cycles often occur during image editing

Technical Basis: Double quantization creates periodic patterns in DCT coefficient histograms

Mathematical Analysis: First quantization: $Q_1(x) = \text{round}\left(\frac{x}{a}\right)$
Second quantization: $Q_2(x) = \text{round}\left(\frac{\text{round}\left(\frac{x}{a}\right) \cdot a}{b}\right)$

Detection Methods: - Histogram analysis for periodic holes/peaks - Fourier analysis of DCT coefficient distributions - Benford's law violation detection in significant digits

1.15 Image editing forensics: inconsistency in light

Physical Principle: Lighting conditions should be consistent across authentic images

Detection Method: Analyze highlight and shadow directions using geometric models

Key Assumptions: - Lambertian surface reflectance - Single dominant light source
- Distant light approximation

Forensic Value: Objects from different sources typically have incompatible lighting

1.16 Image editing forensics: local filtering traces

Operation Detection: Identify specific editing filters through characteristic patterns

Median Filtering: Creates unnatural structural uniformity detectable via statistical analysis
Unsharp Masking: Produces characteristic overshoot artifacts at edges

Forensic Application: These filters are commonly used to hide manipulation traces or enhance doctored regions

1.17 Photo manipulation with inconsistent shadows

1.17.1 Basic assumptions

Analytical Framework: Shadows provide strong physical consistency checks

Shadow Types: - **Cast shadows:** Objects blocking light onto surfaces - **Attached shadows:** Self-shading on object surfaces

Core Principle: Single light source should produce consistent shadow geometry

1.17.2 Cast shadow analysis

1.17.3 Attached shadow analysis

Complementary Techniques: - **Cast shadows:** Define light source position through shadow-object relationships - **Attached shadows:** Determine lighting direction from surface gradient analysis

1.17.4 Mathematical model

Geometric Formulation: Represent shadow constraints as linear inequalities

Constraint System: For each shadow: $n \cdot x - n \cdot p \geq 0$

Complete System: $Nx - P \geq -s$ where slack variables $s \geq 0$ measure consistency violations

Decision Metric: All $s = 0$ indicates physically plausible shadows; any $s > 0$ suggests manipulation

1.18 Traces of image editing: resampling detection

Geometric Transformation Evidence: Resizing/rotation operations leave interpolation patterns

Technical Basis: Resampling creates periodic pixel correlations through interpolation: $y = Hx$

Where interpolation matrix H introduces linearly dependent relationships detectable via Fourier analysis

Forensic Application: Identifies regions that have been geometrically transformed - common in forgeries

1.19 Traces of image editing: contrast enhancement detection

1.19.1 Global contrast enhancement

Histogram Analysis: Contrast stretching creates gaps and peaks in intensity distributions

Frequency Domain: Enhanced images show abnormal high-frequency energy patterns

1.19.2 Local contrast enhancement

Region-Specific Detection: Apply global techniques to image blocks to locate manipulated areas

Implementation: Divide image into blocks (25×25 to 200×200 pixels) and analyze each block independently

Practical Use: Effective for identifying spliced regions with different processing history